

Keep On Track: Element A

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Problem Identification Definition:

Mrs. Stephanie Brennan is a special education teacher at Whitemarsh Elementary School who teaches a LifeSkills class. The LifeSkills class “provides students with intellectual disabilities opportunities to learn functional skills to be more independent — like personal hygiene, communication, social cues, and how to follow a schedule — as well as basic academics.” She needs an assistive device that will aid in cause in effect learning, emotional regulation, attentional control, independence, motor skill development, leisure, cognition, inclusion, and participation. This device needs to be durable enough to withstand daily use, easy to disinfect frequently, securely mounted, deactivatable, ADA compliant, and fully contained. Students must be able to use this continuously and with minimum adult supervision and infrequent maintenance.

Unfortunately, right now barriers impacting access to these types of devices include “low awareness, high costs, limited physical access, inadequate product range, procurement challenges, workforce capacity gaps, inadequate policy, insufficient funding, assistive technology sector fragmentation and sociodemographic obstacles” ([WHO](#)). She believes that a new, cost effective interactive device would be of large benefit to improving the learning environment and capacity of the classroom.

Problem Justification:

Research strongly supports the use of assistive technology in classrooms for students with disabilities. These devices are especially useful for building foundational skills such as cause-and-effect understanding and independence. According to Edyburn (2006), “in the case of students with disabilities in the classroom, assistive technology has the potential to enhance and increase their learning and academic performance,” and Bausch and Hasselbring (2006) further note that “through the use of assistive

technology and digital technologies, students with learning disabilities are able to gain the same benefits as their peers in the classroom.” This is especially important for LifeSkills students, as the Australian Disability Clearinghouse on Education and Training states that children with special learning needs “lack the understanding of both new and complex information and struggle with learning new skills,” making repetitive cause-and-effect learning essential. Additional studies demonstrate that assistive technology has “many advantages in the classroom such as a positive impact on performance for students with disabilities, leading to greater independence and improved quality of life” (Edyburn, 2006). Independence is a key factor for long-term development, as “facilitating their independence in performing the tasks required for involvement in life situations is key to improving both their participation and their development, and therefore, their health and wellbeing” (PMC9915405). Some global health and educational organizations also reinforce this need. UNICEF states that assistive products and technology are intended to give children and youth with disabilities “the opportunity to flourish as others might,” yet “only 1 in 10 people in need actually have access to some assistive products” (UNICEF; PMC9915405). The World Health Organization explains that assistive products “help maintain or improve an individual’s functioning related to cognition, communication, hearing, mobility, self-care and vision, thus enabling their health, well-being, inclusion and participation,” but access remains limited due to “low awareness, high costs, limited physical access, inadequate product range, procurement challenges, workforce capacity gaps, inadequate policy, insufficient funding, assistive technology sector fragmentation and sociodemographic obstacles” (WHO).

These studies and research highlight the proven benefits of assistive technology and the significant accessibility gap that demonstrates the need for cost-effective, accessible solutions, like our device.

Problem Elaboration:

This problem is not only present in Whitemarsh Elementary’s LifeSkills classroom. It is present across all special education and other learning environments.

Assistive learning tools are not only beneficial for students with developmental disabilities, but can also support typically developing students by reinforcing cause-and-effect reasoning, problem-solving skills, and sensory regulation.

Engaging devices that emphasize repetition and independence align with early childhood education goals and can be adapted for use in inclusive classrooms, therapy spaces, and sensory rooms. Outside of that, high costs and limited availability of assistive technology disproportionately affects underfunded schools and families, increasing inequities. Designing a cost-effective and durable assistive device can reduce these disparities and expand access to meaningful learning tools across a variety of educational contexts.

Concern of Primary Stakeholders:

The primary stakeholders affected by this problem include **Life Skills students, classroom teachers, instructional aides, and parents.**

Life Skills students are most directly impacted, as they require safe, engaging tools that promote independence without overwhelming sensory input. The majority of their needs and feedback are measured through observed behavior and educator/parent feedback, due to limited communication abilities.

Teachers and aides are also primary stakeholders. A high level of support is required by the students, so instructional staff spend a significant amount of time managing behavior and engagement. Many of their concerns are about safety, durability, and supervision demands. Therefore, there is a need for a device that students can use independently without constant adult intervention, and still educationally benefit from.

Parents are also primary stakeholders since they witness the effects of the educational tools used in the classroom on their children's behavior and learning at home. As their

children develop new motor skills and cognitive abilities, parents will be interested in how the device helps facilitate growth, and may look into more assistive technology.

Secondary stakeholders include typically developing students in classes with LifeSkills students and school administrators. Peers may benefit socially and educationally from improved behavior and focus, while administrators are positively impacted by cost, durability, safety, and parental recognition. Administrators are also key in the approval of the project as a whole, the resources provided, and the implementation of the device.

Sources:

<https://files.eric.ed.gov/fulltext/EJ1184079.pdf>

<https://exceptionalchildren.org/blog/sett-framework-and-evaluating-assistive-technology-remotely?srsltid=AfmBOoqpufFz4bX3RGcf91sqbmXx8ZHY2CGmXK7CoB2DdZ1lur5m0Hr5>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9915405/>

<https://www.who.int/news-room/fact-sheets/detail/assistive-technology>

<https://www.adcet.edu.au/inclusive-teaching/specific-disabilities/intellectual-disability>